

Statistical Concepts Used in the Medical Insurance Cost Analysis Project

1. Descriptive Statistics

- **Mean:** Arithmetic average, $\bar{x} = \sum x/n$.
- **Median:** Middle observation after sorting; less affected by extreme values.
- **Standard deviation:** Measures spread around the mean.
- **IQR:** $Q3 - Q1$; spread of the middle 50% of observations.
- **Skewness:** Measures asymmetry. Positive values indicate a longer right tail.
- **Kurtosis:** Describes tail heaviness/peakedness; the project's output reports excess kurtosis.

2. Hypothesis Testing

- Hypothesis testing uses sample evidence to assess a null hypothesis (H_0) against an alternative hypothesis (H_1).
- **Significance level:** $\alpha = 0.05$ in this project.
- **p-value:** Evidence against H_0 . When $p < 0.05$, the project rejects H_0 and treats the result as statistically significant.

3. Shapiro–Wilk Normality Test

- **Purpose:** Tests whether sample data are consistent with a normal distribution.
- **H_0 :** Data are normally distributed.
- **H_1 :** Data are not normally distributed.
- Decision rule: $p < 0.05 \rightarrow$ reject H_0 .

4. Levene's Test

- **Purpose:** Tests equality of variances between groups.
- **H_0 :** Group variances are equal.
- **H_1 :** Group variances are not equal.
- It is useful when checking assumptions for a standard two-sample t-test.

5. Two sample t-test

- Compares the means of two independent groups when the relevant parametric assumptions are reasonable.
- **H_0 :** The two population means are equal.
- **H_1 :** The means are different.

6. Mann–Whitney U Test

- A non-parametric alternative to the independent two-sample t-test.
- In this project, it was used for smoker vs non-smoker charges because the Shapiro–Wilk tests showed strong departures from normality.

7. One-Way ANOVA

- Tests whether the means of three or more independent groups are equal.
- **H0**: All group means are equal.
- **H1**: At least one group mean differs.
- The **F-statistic** compares between-group variation with within-group variation.
- In this project, ANOVA compares average insurance charges across the four regions.

8. Multiple Linear Regression

- Models a continuous response using several explanatory variables.
- Project model: **charges ~ age + bmi + children + sex + smoker + region**.
- General form: $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \varepsilon$.
- **Coefficient** β_j : expected change in Y for a one-unit change in X_j , holding other predictors constant.

9. Ordinary Least Squares (OLS)

- OLS estimates regression coefficients by minimizing the sum of squared residuals.

10. Coefficients, p-values and 95% Confidence Intervals

- **Coefficient**: Estimated association between a predictor and charges, controlling for the other predictors.
- **Coefficient p-value**: Tests whether the coefficient differs significantly from zero.
- **95% confidence interval**: Gives a range of plausible values for the population coefficient.

11. R^2 and Adjusted R^2

- **R^2** : Proportion of variation in charges explained by the predictors.
- **Adjusted R^2** : Adjusts R^2 for the number of predictors and is useful for comparing models with different predictor counts.
- Project result: $R^2 \approx 0.751$ and adjusted $R^2 \approx 0.749$.

12. Overall F-Test for Regression

- Tests whether the regression model has explanatory power beyond an intercept-only model.
- **H0**: All slope coefficients are zero.
- **H1**: At least one slope coefficient is non-zero.

13. Residuals

- A residual is the difference between the observed and fitted response:

14. Regression Assumptions

- **Linearity:** The systematic relationship is adequately represented by the linear model.
- **Independence:** Observations/errors are independent.
- **Homoscedasticity:** Error variance is approximately constant across fitted values.
- **Residual normality:** Residuals are approximately normal when normal-based inference is required.

15. Jarque–Bera and Omnibus Tests

- **Jarque–Bera:** Uses residual skewness and kurtosis to assess normality.
- **Omnibus:** General test of departure from normality.
- The project's residual diagnostics show strong evidence that residuals are not normally distributed.

16. Q–Q Plot

- Compares observed residual quantiles with theoretical normal quantiles. Points close to the reference line suggest approximate normality; systematic deviations indicate departures.

17. Variance Inflation Factor (VIF)

- VIF diagnoses multicollinearity among predictors. High values indicate that a predictor is strongly related to the other predictors.
- The project's VIF values were approximately 1.00–1.65, indicating no serious multicollinearity.